

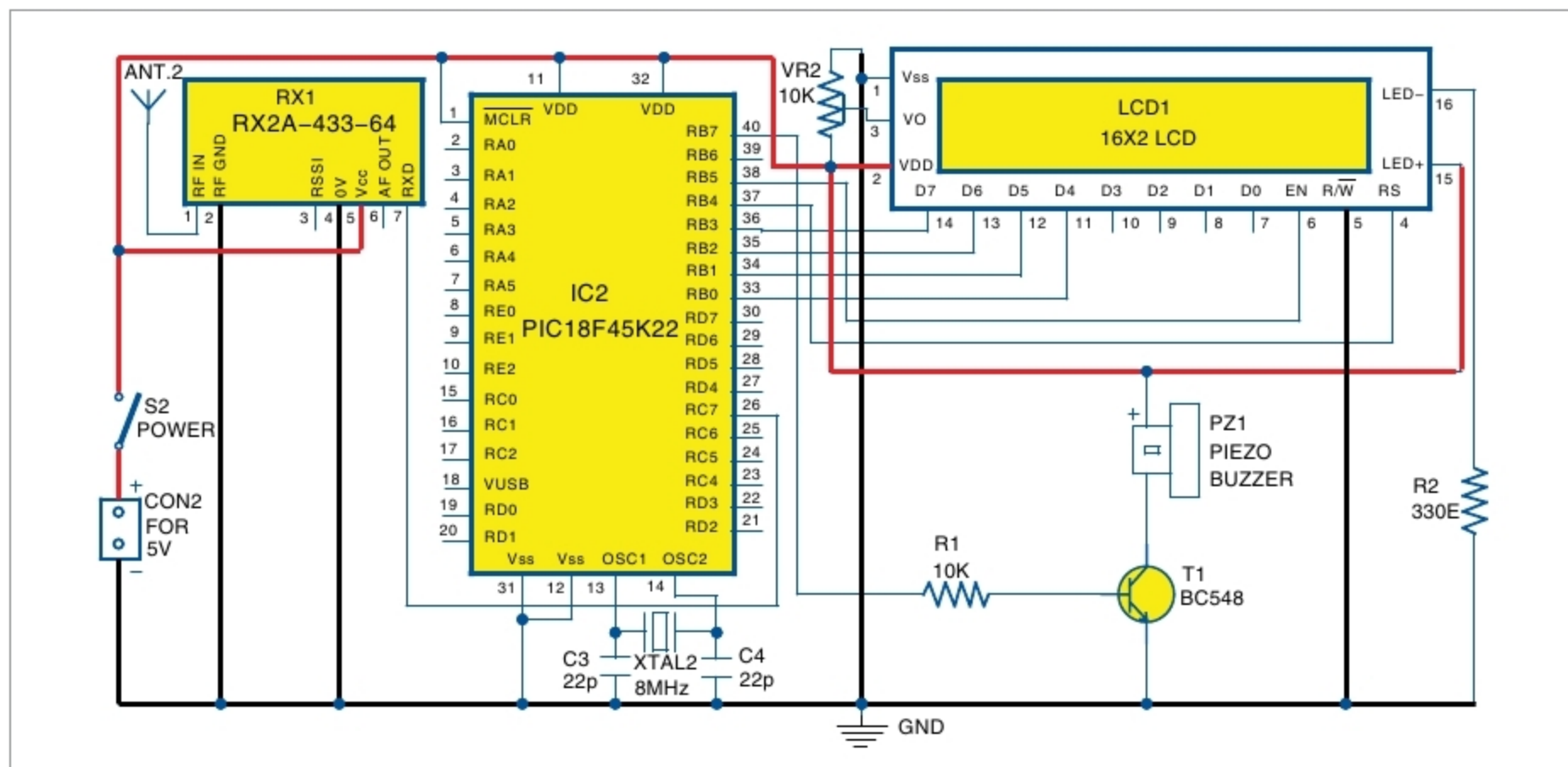


Fig. 7: Circuit diagram of monitoring circuit

heater is used. The sensor provides an analogue output voltage proportional to ambient gas concentration.

Design of wireless gas detection and monitoring systems

The block diagram of the designed system is shown in Fig. 2. The system consists of two parts: gas detection stations and gas monitoring station. A gas detection station consists of a semiconductor gas sensor and an MCU that receives signals from the gas sensor and activates a suitable transmitter module. This detection or transmitter station sends a message to the monitoring station along with a unique station ID to alert the user about the gas leakage along with its location.

In this design, four switches are used, which makes it possible to have up to 16 detection stations. Their unique IDs range from 0 to 15, selected using switches.

Circuit and working

Both detection and monitoring stations use low-cost 8-bit medium-performance PICF family of MCUs. Hardware details of each station are given below.

Detection and transmitter stations. MQ-6 semiconductor LPG gas sensor (Fig. 3) is used in the design, since it is highly sensitive, low cost

and easy to use. LPG concentrations above 1000ppm are accepted to be toxic and should be avoided. The system has been designed to activate if LPG concentration is above 1000ppm.

MQ-6 pins are organised in groups of three pins on each side of the sensor, where pins 2 and 5 are heater pins, and pins 1 and 3 and pins 4 and 6 are sensor circuit pins. As shown in Fig. 1, load resistor R_L is connected in series with sensor resistance R_s to convert resistance R_s to voltage. Fig. 4 shows the sensor output voltage when a 4.7-kilo-ohm load resistor is used.

The circuit diagram of the gas detection station is shown in Fig. 5. It is built around an MQ* series LPG gas sensor, PIC18F2550 MCU (IC1), TX2H-433-64-5V 433MHz transmitter module (TX1), 4-way DIP switch (DIP1) and a few other components. MQ* LPG sensor may be MQ-3, MQ-4, MQ-6, MQ-7 or MQ-8.

PIC18F2550 is used in the design with an external 8MHz crystal. PLL clock multiplier is used to increase the operating clock frequency from 8MHz to 48MHz. Sensor output is connected to a 10-kilo-ohm variable resistor, which feeds into analogue input channel RA1 of the MCU.

Sensitivity of the device can easily be adjusted via this potentiometer. Whenever a gas leakage is detected, a

message is sent to the monitoring station together with the unique station ID. Every station is assigned a unique ID by using the switches connected to Port B.

Depending upon applications, distance between detection stations and monitoring station can be hundreds of metres. In a large caravan site, this distance can even be a few kilometres. For this reason, RF transmitter and receiver modules are used in the design. Radiometrix (www.radiometrix.co.uk) TX2H transmitter module (Fig. 6) is used in the design. It is a 433.92MHz UHF FM transmitter module with 25mW power output, having a typical range of 500 metres, consuming only 23mA current.

The module is powered from +5V supply and is connected to pin 17 or UART (TX) of the MCU. A helical loop or whip antenna can be used with the module. Length of the whip antenna should be 16.4cm from RF pin. Pin configuration of TX2H is shown in Table I. Gas detection stations can be powered from +5V mains adaptors, usually used to charge cellphones.

Receiver and monitoring station. Fig. 7 shows the circuit diagram of the monitoring station. It is built around 433MHz receiver module RX2A-433-64 (RX1), PIC18F45K22 MCU (IC2), 16x2 LCD (LCD1), transistor BC548